Ed (Ehsan) Shah Hosseini

Optical Network Development Engineer at AWS, US Citizen

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Summary

Silicon photonics engineer with 20 years in CMOS-compatible photonic integration, spanning datacenter optics, consumer sensing, and defense applications. 53 granted US patents in optical phased arrays and LiDAR, and publications in Nature and Nature Communications (5,000+ citations). Principal Investigator on $3.9M+ in DOE and DoD SBIRs at Analog Photonics, and co-author on the founding proposal for AIM Photonics, the $600M+ Manufacturing USA institute. Currently building 800G/1.6T intrarack optical links at AWS; completed Andrew Ng's Deep Learning Specialization.

Experience

2026 **Amazon Web Services (AWS)**, Optical Network Development Engineer, Cupertino, CA

- Design and validate 800G intrarack optical links (100G/lane, OSFP) deployed in AWS datacenter networks
- Implement in-band 1PPS delivery of GPS-sourced timing from TOR to NIC, holding jitter under 10ns over existing high-speed optical links
- Build an asynchronous side channel on the same optical link for telemetry and firmware updates, targeting 1-4 Mb/s alongside 1PPS delivery
- Define architecture and vendor requirements for next-generation 1.6T links (200G/lane)

20–26 **Apple Camera Technology Group**, Silicon Photonics Lead, Cupertino, CA

- Coordinated cross-discipline team (VCSEL, TIA, driver, optics, applications) on R&D for SMI-based OCT consumer sensing; set system requirements (SNR, noise budget) and made component-level design decisions
- Led R&D on coherent laser Doppler imaging for wearable health sensing
- Built beam-scanning rig for 3D OCT capture and set up high-density SMI VCSEL array with coverglass backreflection compensation
- Designed FMCW chirp pre- and post-processing to linearize all lasers across the SMI VCSEL array
- Led MATLAB platform for end-to-end coherent sensing system simulation

14–20 **Analog Photonics**, Director of Advanced Projects and Photonics Integration

- Co-authored winning proposal for AIM Photonics, the $600M+ Manufacturing USA institute for integrated photonics ($110M AFRL, $250M New York State, $245M+ public/private match), 2015
- Primary author of the first industry PDK at AIM Photonics, the foundry design kit used by external silicon-photonics customers; sustained $1M+ annual AP funding through the institute
- Principal Investigator on DOE SBIR program for photonic memory controller module ($2.8M across Phases I, II, IIB): 30 Gb/s MZI-based athermal optical links and microring WDM toward 800GE on-board optics; multi-company consortium with Freedom Photonics and PLCC
- Co-developed 400G-FR4 silicon photonics transceiver chipsets for CPO, OBO, and pluggable form factors (94 Gb/s/mm density, <10 pJ/bit, -5.4 dBm OMA at KP4 pre-FEC BER); demonstrated at OFC 2019
- Invented two LiDAR architectures: WDM LiDAR combining multi-wavelength sources with optical phased array, and switched-array LiDAR using curved-focal-plane emitters with log2(N) control signals; demonstrated chip-scale in silicon
- Principal Investigator on five additional SBIRs ($1M+ combined) across Army, Navy, and DoD: hybrid MOPA laser at 1550 nm (>1 W output, >1 GHz/µs chirp) with OPA-based coherent combining, UAV/tactical free-space optical comms, 3D optical sensors for Future Vertical Lift, and silicon photonics optogenetic transceivers

11–14 **MIT Photonics Microsystems Group**, Post-doctoral Associate

- Designed, fabricated, and demonstrated DFB and DBR erbium-doped lasers integrated into a CMOS-compatible silicon photonics fab, enabling on-chip light sources without III-V bonding
- Developed germanium-on-silicon photodetectors from process up at 1550 nm: ∼1 A/W responsivity, 50 GHz bandwidth, nA-scale dark current
- Contributed to large-scale 2D nanophotonic optical phased array demonstration (Nature, 2013)
- Contributed to ultralow-power athermal silicon modulator demonstration (Nature Communications, 2014)
- Designed silicon photonics device library: PN-junction MZI modulators, microdisk and microring filters and modulators, edge couplers, and directional grating couplers

Education

03–11 **Georgia Institute of Technology (Georgia Tech), Atlanta, GA**
MS and PhD, Electrical Engineering (Photonics) | Minor: Physics
Thesis: "High Quality Integrated Silicon Nitride Nanophotonic Structures for Visible Light Applications."

99–03 **Aryamehr (Sharif) University of Technology, Tehran, Iran**
BSc, Electrical Engineering | Top 10 in EE, Emphasis: Fields & Waves in Communications

Continuing Education

23–25 **DeepLearning.AI (Coursera)**
Deep Learning Specialization (Andrew Ng)
Machine Learning Specialization (Andrew Ng)

Selected Patents

Analog Photonics (2019–2024)

1. "Multi-static coherent LiDAR," US Patent 12,050,265 B2, 2024.
2. "Optical phased array light steering," US Patent 12,085,833 B2, 2024.
3. "Large scale steerable coherent optical switched arrays," Patent Application US20240329324A1, 2024.
4. "Aberration correction of optical phased arrays," US Patent 12,517,351 B2, 2026.
5. "Photonic die alignment," US Patent 11,693,196 B2, 2023.
6. "Photonics fabrication process performance improvement," US Patent 11,762,146 B2, 2023.
7. "Tunable optical structures," US Patent 11,269,236 B2, 2022.
8. "Optical switching using spatially distributed phase shifters," US Patent 11,422,431 B2, 2022.
9. "Speckle reduction in photonic phased arrays," US Patent 11,353,769 B2, 2022.
10. "Two dimensional optical phased arrays using edge-coupled integrated circuits," Patent Application WO2022051453A1, 2022.
11. "Array-based free-space optical communication links," US Patent 11,196,486 B2, 2021.
12. "Wavelength division multiplexed lidar," US Patent 11,061,140 B2, 2021.
13. "Optical phase shifter device," US Patent 10,809,591 B2, 2020.
14. "Large scale steerable coherent optical switched arrays," US Patent 10,338,321 B2, 2019.

Apple (2023–2024)

1. Depth sensing using multiple coherent emitters with interleaved VCSEL/detector arrays (WO2024177716A1, 2024)
2. Coherent optical waveguide for wearable biometric sensing (US20240057896A1, 2024)
3. "LiDAR array with vertically-coupled transceivers," Patent Application US20230366986A1, 2023.

MIT (2019)

1. "Photonic devices and methods of using and making photonic devices," US Patent 10,461,489 B2, 2019.
2. "Apparatus, systems, and methods for waveguide-coupled resonant photon detection," Patent Application US20170219776A1, 2017.

Selected Publications

Nature and High-Impact Journals

1. J. Sun, ..., E. S. Hosseini, "Large-scale nanophotonic phased array," *Nature*, 2013. (1,670 citations)
2. E. Timurdogan, ..., E. Shah Hosseini, "An ultralow power athermal silicon modulator," *Nature Communications*, 2014. (569 citations)

Silicon Photonics and Integrated Optics

1. E. Shah Hosseini, ..., "High quality planar silicon nitride microdisk resonators for integrated photonics in the visible wavelength range," *Optics Express*, 2009. (219 citations)
2. J. Sun, ..., E. S. Hosseini, "Large-scale silicon photonic circuits for optical phased arrays," *IEEE JSTQE*, 2013. (210 citations)
3. E. Shah Hosseini, ..., "Systematic design and fabrication of high-Q single-mode pulley-coupled planar silicon nitride microdisk resonators at visible wavelengths," *Optics Express*, 2010. (190 citations)
4. A. H. Atabaki, E. S. Hosseini, ..., "Optimization of metallic microheaters for high-speed reconfigurable silicon photonics," *Optics Express*, 2010. (158 citations)
5. J. Sun, E. S. Hosseini, ..., "Two-dimensional apodized silicon photonic phased arrays," *Optics Letters*, 2014. (128 citations)
6. J. D. B. Bradley, E. S. Hosseini, ..., "Monolithic erbium- and ytterbium-doped microring lasers on silicon chips," *Optics Express*, 2014. (117 citations)
7. Purnawirman, ..., E. S. Hosseini, "C- and L-band erbium-doped waveguide lasers with wafer-scale silicon nitride cavities," *Optics Letters*, 2013. (99 citations)
8. E. S. Hosseini, ..., "CMOS-compatible 75 mW erbium-doped distributed feedback laser," *Optics Letters*, 2014. (88 citations)
9. Purnawirman, ..., E. S. Hosseini, "Ultra-narrow-linewidth Al_2O_3:Er^{3+} lasers with a wavelength-insensitive waveguide design on a wafer-scale silicon nitride platform," *Optics Express*, 2017. (56 citations)
10. Z. Su, ..., E. S. Hosseini, "Four-port integrated polarizing beam splitter," *Optics Letters*, 2014. (56 citations)

Foundry and Systems (2018)

1. E. Timurdogan, ..., E. S. Hosseini, "AIM process design kit (AIMPDKv2.0): Silicon photonics passive and active component libraries on a 300mm wafer,"
OFC, 2018. (54 citations)
2. C. V. Poulton, ..., E. Hosseini, "High-performance integrated optical phased arrays for chip-scale beam steering and LiDAR," *CLEO*, 2018. (50 citations)